Fiscal Year:	FY 2026	**Task Last Updated:**	FY 02/24/2026
PI Name:	Sanford, Larry Ph.D.		
Project Title:	Impact of Inflight Stress and Sleep Disturbances on Brain Function, Neural Communication, and Inflammation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	10/01/2019	**End Date:**	11/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2025 per L. Juliette (Ed., 8/5/25) NOTE: End date changed to 09/30/2025 per L. Juliette (Ed., 5/8/22)		
Key Personnel Changes/Previous PI:	July 2021 report: Shao-Hui Chuang and Richard Ciavarra are no longer working on the project. July 2020 report: None. August 2024 report: None. February 2026 report: None.		
COI Name (Institution):	Britten, Richard Ph.D. (Eastern Virginia Medical School) Wellman, Laurie Ph.D. (Eastern Virginia Medical School) Yoon, Hargsoon Ph.D. (Norfolk State University)		
Grant/Contract No.:	80NSSC19K1582		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts may experience several physical and psychological challenges including prolonged exposure to microgravity, potential social isolation (SI), and predicted exposure to ~13 cGy/Yr Space Radiation (SR). Each of these stressors can have deleterious effects on physical and mental health as well as negatively impact mission performance. Each also can disturb sleep, which can further exacerbate the effects of stress. Determining how astronauts respond to the single and compound effects of mission related stressors is crucial for understanding and mitigating factors that could impede mission success. However, examining their effects will be complex because they have different time courses and durations. SR exposure will be episodic whereas microgravity will be constant and SI and sleep disturbances can be temporary or prolonged. Their effects may also be additive or synergistic, and may vary with sex and with individual differences in stress resilience and vulnerability. Therefore, our goal in this project is to determine the single and synergistic effects of inflight stress and SR on mission related biological and functional challenges that astronauts may face. We will determine how order of experience impacts the effects of inflight stressors and determine sex and individual differences in stress resiliency using an established Wistar rat model. We will use established ground-based models of microgravity (hindlimb unloading), SI, and SR. We will use telemetry to record sleep and to assess temperature increases as an index of the stress response. We will use plasma levels of brain derived neurotropic factor to predict individual differences in stress responses. We will assess fear learning and extinction to determine how chronic inflight stressors impact ability to respond to an additional acute stressor. We will analyze food entrained, anticipatory activity to assess potential impairments in temporal perception. We will determine the effects of inflight stress and SR on sensorimotor function and recognition memory using behavioral tests. We will use local field potential recordings in the amygdala, medial prefrontal cortex, and hippocampus to determine whether stressors and SR alter the coordinated neural activity underlying fear learning and extinction, and we will determine whether there are structural changes within connections between regions. We will use chemokine/cytokine arrays and Ingenuity Pathway Analysis to determine the effect of stressors and SR on the immune system. Collectively these studies will assess functional, mechanistic, and immune consequences of several stressors inherent in space exploration. The data they provide will aid in risk assessment, and may help to develop countermeasures based on individual differences in ability to withstand stress and/or targeting immune pathways that may confer protection against the deleterious effects of multiple stressors that astronauts will face.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies will increase understanding of how individual differences in resilience and vulnerability factor in determining lasting outcomes of stressful experiences. They will also determine the role that stress-induced alterations in sleep play in mediating the outcomes of stress. The results will have importance for understanding how individuals cope with stress on Earth and on missions in space.
Task Progress:	Task Progress Our goal in this project was to determine the impact and potential synergistic effects that multiple inflight stressors and exposure to space radiation (SR; 15cGy GCRsim) can have on biological systems and mission related performance. Outcomes that we examined include sensorimotor function and balance (adhesive removal task and balance beam performance), recognition memory (novel object recognition task), time perception (anticipatory activity), fear memory, and fear memory related neural communication. We are also examining the relationship of altered performance to the effects of inflight stressors on sleep, and how responses to stressors vary in resilient (Res) animals that maintain sleep after experiencing stress and vulnerable (Vul) animals that have reduced sleep after experiencing stress. This was the final year of this project. We completed a planned cross-species validation study in mice aimed at determining the effects of controllable and uncontrollable stress on mice that had experienced SR. Unfortunately, as with hindlimb unloading in female rats, irradiated mice showed high levels of mortality (~75%), which precluded continuation of surgery and subsequent sleep recording. We were able to perform a preliminary analysis of freezing behavior in our controllable/uncontrollable footshock paradigm. C57BL/6 male mice (18 wks old) were exposed to either SR or served as non-irradiated controls (Shams) at Brookhaven National Laboratory (BNL) and were then shipped to Eastern Virginia Medical School (EVMS) for study. Three months post-irradiation, mice were assigned to either controllable (escapable) shock (ES-Sham, n=6; ES-SR, n=6) or uncontrollable (inescapable) shock (IS-Sham n=6; IS-SR, n=6). Over two consecutive days all mice were subjected to shock training (20-foot shocks, 0.5mV, up to 5 seconds) delivered through a metal grid floor every 60 seconds. ES mice could stop the shock by moving into the other chamber; IS mice could not (their shock was yoked to the ES chamber). Seven days after shock training (ST), the mice were exposed for a 30-minute period in the same chamber without shocks, to assess contextual fear memory. Twenty-one days after shock training (14 days after CTX), the mice were exposed to the chamber for a second 30-minute period without shocks to assess potential to extinction of freezing. Freezing did not significantly differ between ES and IS Sham mice across ST1, ST2, freezing during context (CTX), and freezing during extinction (EXT) conditions. This is consistent with prior studies in our lab. IS-SR rats showed increased freezing on the CTX day compared to ES-IR mice. ES-SR mice showed reduced freezing compared to IS-IR mice and compared to non-irradiated ES- and IS-Sham mice. Reduced freezing in ES-IR mice throughout the EXT day trial suggests that these mice experienced more rapid extinction than other groups. Reduced freezing in ES-IR mice during CTX and EXT suggests that controllability can alter fear learning after exposure to SR. These data demonstrate that controllable and uncontrollable stress can differentially impact behavior in irradiated mice. We also collected tissue to assess immune and stress markers to determine how SR impacts neuroinflammatory and stress responses associated with controllable and uncontrollable stress, and memories of controllable and uncontrollable stress. We began a study examining the effect of SR given to rats that had been in social isolation (SI) for two weeks prior to irradiation and continuing afterwards. Male (n=16) and female (n=16) rats that were under SI received SR at BNL and returned to EVMS for study. Unfortunately, due to an abrupt termination of funding, this study was cut short and only balance beam studies were performed. Brain tissue also was collected and stored for potential future analysis. Analysis for the mouse study was also halted. With respect to the overall project, we completed studies on SR alone, SI alone, and SI beginning after SR in both males and females. Data collected include adhesive removal task and balance beam performance, novel object recognition task, anticipatory activity, sleep, and indices of fear memory (conditioned changes in sleep, freezing, and stress-induced

	hyperthermia). The data are being prepared for publication. Notable results for the project overall include findings that sleep resilience to stress predicts resilience to other effects of SR in male rats. Vul rats had increased BBB damage and heightened immune response compared to Res. However, freezing did not differ in Res and Vul male rats, regardless of treatment. By comparison, sham females had higher level of freezing than sham males and SR did not increase freezing in females. SR increased freezing in males to the level seen in females. Both SR and SI impacted sleep and EEG spectra with changes indicating less restorative sleep with higher arousal indices during sleep. SI produced performance differences in sensorimotor function (balance beam, adhesive removal), even compared to single-housed animals. These changes involve both sensorimotor function and emotion. Lastly, SI after SR attenuated some effects of SR; e.g., SR animals had a 1.5 to 5.6-fold enlargement of the lateral ventricles compared to any other group. These effects were attenuated when SR was combined with SI. In summary, our data to date demonstrate both single and synergistic effects of SI and SR on several behaviors and sleep, significant differences between males and females, and individual differences in resilience in the effects of SR on behavior and sleep. They provide a framework for understanding how SI and SR can impact multiple, interacting systems (i.e., fear memory, stress, and sleep) as well as sensorimotor performance.
Bibliography Type:	Description: (Last Updated: 03/16/2026)
Abstracts for Journals and Proceedings	Sanford LD, Wellman LL, Luyo ZNM, Heerbrandt RS, Boden AF, Singh N, Britten RA. "Effects of spaceflight hazards on stress responses, conditioned fear, and sleep in stress resilient and vulnerable rats." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Luyo ZNM, Heerbrandt RS, Boden AF, Britten RA, Wellman LL, Sanford LD. "Differential effects of spaceflight hazards on neuroimmune cell morphology." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstract. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Articles in Peer-reviewed Journals	Adkins AM, Luyo ZNM, Boden AF, Heerbrandt RS, Britten RA, Wellman LL, Sanford LD. "Neuroinflammatory signaling and immune cell infiltration differ in brains of rats exposed to space radiation and social isolation." Life. 2025 May 6;15(5):747. https://doi.org/10.3390/life15050747 ; PMID: 40430175; PMCID: PMC12113652 , May-2025
Articles in Peer-reviewed Journals	Adkins AM, Boden AF, Singh N, Luyo ZNM, Britten RA, Wellman LL, Sanford LD. "Effects of 15 cGy GCRsim space radiation on conditioned fear and stress responses in stress resilient and vulnerable rats: Implications for astronaut sleep, health and mission performance." Radiat Res. 2026 Jan 20;205(3):259-272. https://doi.org/10.1667/RADE-25-00092.1 ; PMID: 41679740 , Jan-2026
Articles in Peer-reviewed Journals	Sanford LD, Adkins AM, Boden AF, Luyo ZZNM, Singh N, Britten RA, Wellman LL. "Effects of social isolation alone and combined with 15cGy GCRsim space radiation on fear conditioned behavioral, stress, and sleep responses." Life Sci Space Res. 2026 Jan:48:132-142. https://doi.org/10.1016/j.lssr.2025.10.001 ; PMID: 41565393 , Jan-2026